

WENRUI HUANG

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EDUCATION	Brown University <i>Ph.D. in Mathematics</i> • Advisor: Benoît Pausader	Providence, RI 2022 - 2027 (<i>expected</i>)
	University of Science and Technology of China (USTC) <i>B.S. in Mathematics</i> • Advisor: Yong Wei • Thesis: Mean curvature flow with surgery and its applications	Hefei, China 2018 - 2022
RESEARCH INTERESTS	Partial differential equations, nonlinear dispersive PDEs and kinetic equations, stability and long-time behavior.	
PAPERS AND PREPRINTS	1. A common structure for modified scattering: Vlasov-Riesz, Hartree, and their coupling , with M. Xie. arXiv:2610.02643. 2. Asymptotic behavior of the one-dimensional Vlasov-Poisson system with a repulsive harmonic potential , arXiv:2610.02551. 3. Long-Time Dynamics of Vlasov-Hartree Systems across the Coulomb threshold , with M. Xie. arXiv:2608.04402. 4. Scattering map for the Vlasov-Poisson system with a repulsive harmonic potential , with H. Kwon. arXiv:2602.21344. 5. Stability and instability of small BGK waves , with D. Bian, E. Grenier, and B. Pausader. arXiv:2601.10030. 6. The Vlasov-Poisson System with a Perfectly Conducting Wall: Convex Domains , with B. Pausader and M. Suzuki. Commun. Math. Phys. 407, 219 (2026). 7. Modified wave operators for the Hartree equation with repulsive Coulomb potential , arXiv:2411.02737. 8. Scattering of the Vlasov-Riesz system in three dimensions , with H. Kwon. Annales Henri Poincaré (2026). arXiv:2407.16919.	
TALKS AND PRESENTATIONS	• Analysis Seminar, Yale University • GLESPA Seminar, Brown University • Poster Session, Workshop on Kinetic Theory and Fluids, University of Wisconsin–Madison • Brown–Yale PDE Seminar • GLESPA Seminar, Brown University • Brown–Yale PDE Seminar	Sep. 2026 Apr. 2026 Mar. 2025 Apr. 2024 Apr. 2024 Nov. 2023
TEACHING	Brown University • TA, MATH 0180 Calculus III, Fall 2026 • TA, MATH 0190 Calculus II (Physics/Engineering), Fall 2025 • TA, MATH 0180 Calculus III, Spring 2024 • TA, MATH 0180 Calculus III, Fall 2023 • TA, MATH 0100 Calculus II, Spring 2023 University of Science and Technology of China • TA, Riemannian Geometry, Spring 2022 • TA, Differential Geometry, Fall 2021 • TA, Topology, Spring 2021 and Spring 2022	